

Product Forge — Pipeline Stages & Agents

Reference

> **Generated** from `pipeline-definition.json` + `agents/\*.agent.json`
(`scripts/gen\_pipeline\_reference.py`). Do not hand-edit — re-run the generator.

39 stages across the phases below. Every stage's canonical artifact is
`artifacts/<stage>/<agent>-output.md`; extra formats (.html/.pdf/.xlsx) are derived per
`config/artifact-formats.json`. Stage directories use human-readable names (`<id> - <Name>`, e.g. `1 - Design`) per `config/artifact-paths.json`; each holds a `\_stage.json` and the tree has an
`artifacts/INDEX.md` (3a).

Phases

Phase	Stages
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P1 Ideation & Discovery	0, 0a
P2 Business, Market & Monetization	0b, 0c, 0d, 0e
P3 Design	1, 1a, 1b, 1c, 1d
P4 Architecture	2, 3, 3a
P5 Implementation	4-0, 4a, 4a-vqa, 4b, 4b-vqa, 4c, 4c-vqa, 4d, 4d-vqa, 4e, 4e-vqa, 4f, 4f-vqa
P6 Verification	5, 6, 7
P7 Delivery	8, 9, 10, 10a, 11, 12
P8 Operate, Grow & Engage	13, 13a, 13b

Stages

Stage	Disp	Name	Phase	Agent(s)	Role	Artifact(s)	Prev	Next	Opt	Gate	
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0	S1.1	Ideation	P1 Ideation & Discovery	`ideation`	Discovery process — explores, challenges, discovers what user actually needs	`artifacts/0/ideation-output.md` `docs/product-plan.md`	—	→ 0a			
0a	S1.2	Discovery	P1 Ideation & Discovery	`discovery`	Run structured discovery to shape the product.	`artifacts/0a/discovery-output.md` `docs/idea-refined.md` `docs/domain-analysis.md` `docs/stakeholder-map.md` `docs/user-personas.md` `discovery-panel.json` `discovery-questions.json`	0 → 0b				
0b	S2.1	Business & Product Definition	P2 Business, Market & Monetization	`product-owner`	Works WITH the ideation and discovery agents to distill the idea into product goals, vision, scope, personas and success metrics	`artifacts/0b/product-owner-output.md` `product-plan.json`	0a → 0c		yes	AG-business	
0c	S2.2	Market, Competition & Positioning	P2 Business, Market & Monetization	`researcher`	Conducts thorough research on topics using web search, academic sources, and industry knowledge.	`artifacts/0c/researcher-output.md` `marketing-strategy/positioning.md`	0b → 0d		yes	AG-market	
0d	S2.3	Monetization & Unit Economics	P2 Business, Market & Monetization			`pricing-strategist`	Pricing & monetization lead	`artifacts/0d/pricing-strategist-output.md`	0c → 0e	yes	AG-monetization
0e	S2.4	Go-to-Market & Customer Journey	P2 Business, Market & Monetization	`marketing`	Marketing	`artifacts/0e/marketing-output.md` `marketing-strategy/gtm-strategy.md` `marketing-strategy/campaign-plan.md`	0d → 1		yes	AG-gtm	
1	S3.1	Design	P3 Design	`design`	Extracts formal requirements and produces a design doc from the product plan	`artifacts/1/design-output.md` `docs/design.md` `docs/requirements.md`					

`artifacts/1/features/F-\*.functional.md` | 0e → 1a | | AG-scope-change |
****1a****	S3.2	Product Design Spec	P3 Design	`product-design-spec`	Produce a structured product design specification.	`artifacts/1a/product-design-spec-output.md` `design-spec.json`	1 → 1b		
****1b****	S3.3	Design Review	P3 Design	`design\_critic`	Review the design for quality and completeness.	`artifacts/1b/design\_critic-output.md`	1a → 1c		
****1c****	S3.4	UX/IA Review	P3 Design	`ux-ia`	Define UX and information architecture + design tokens.	`artifacts/1c/ux-ia-output.md`	1b → 1d		
****1d****	S3.5	Research	P3 Design	`researcher`	Conducts thorough research on topics using web search, academic sources, and industry knowledge.	`artifacts/1d/researcher-output.md`	1c → 2		
****2****	S4.1	Architect	P4 Architecture	`architect`	Chooses the tech stack and produces the architecture document (ADRs) from the requirements and design.	`artifacts/2/architect-output.md` `docs/architecture.md`	1d → 3		AG-architecture
****3****	S4.2	Refine Requirements	P4 Architecture	`orchestrator`	Project Coordinator (judgment plane)	`artifacts/3/orchestrator-output.md`	2 → 3a		
****3a****	S4.3	QA Spec Review	P4 Architecture	`validate`	Senior QA Engineer	`artifacts/3a/validate-output.md`	3 → 4-0		
****4-0****	S5.0	Skeleton	P5 Implementation	`implement`, `devops`	Builds the code from the approved design, architecture, and requirements / Devops	`artifacts/4-0/implement-output.md` `artifacts/4-0/devops-output.md`	3a → 4a		
****4a****	S5.1	Implementation Iteration 1	P5 Implementation	`implement`, `devops`, `code-review`, `validate`	Builds the code from the approved design, architecture, and requirements / Devops / Code Review agent (Stage 5) / Senior QA Engineer	`artifacts/4a/implement-output.md` `artifacts/4a/devops-output.md` `artifacts/4a/code-review-output.md` `artifacts/4a/validate-output.md`	4-0 → 4a-vqa		
****4a-vqa****	S5.1a	Visual QA (Iteration 1)	P5 Implementation	`visual\_qa`	Verify rendered UI against the design spec	`artifacts/4a-vqa/visual\_qa-output.md`	4a → 4b		
****4b****	S5.2	Implementation Iteration 2	P5 Implementation	`implement`, `devops`, `code-review`, `validate`	Builds the code from the approved design, architecture, and requirements / Devops / Code Review agent (Stage 5) / Senior QA Engineer	`artifacts/4b/implement-output.md` `artifacts/4b/devops-output.md` `artifacts/4b/code-review-output.md` `artifacts/4b/validate-output.md`	4a-vqa → 4b-vqa		
****4b-vqa****	S5.2a	Visual QA (Iteration 2)	P5 Implementation	`visual\_qa`	Verify rendered UI against the design spec	`artifacts/4b-vqa/visual\_qa-output.md`	4b → 4c		
****4c****	S5.3	Implementation Iteration 3	P5 Implementation	`implement`, `devops`, `code-review`, `validate`	Builds the code from the approved design, architecture, and requirements / Devops / Code Review agent (Stage 5) / Senior QA Engineer	`artifacts/4c/implement-output.md` `artifacts/4c/devops-output.md` `artifacts/4c/code-review-output.md` `artifacts/4c/validate-output.md`	4b-vqa → 4c-vqa		
****4c-vqa****	S5.3a	Visual QA (Iteration 3)	P5 Implementation	`visual\_qa`	Verify rendered UI against the design spec	`artifacts/4c-vqa/visual\_qa-output.md`	4c → 4d		
****4d****	S5.4	Implementation Iteration 4	P5 Implementation	`implement`, `devops`, `code-review`, `validate`	Builds the code from the approved design, architecture, and requirements / Devops / Code Review agent (Stage 5) / Senior QA Engineer	`artifacts/4d/implement-output.md` `artifacts/4d/devops-output.md` `artifacts/4d/code-review-output.md` `artifacts/4d/validate-output.md`	4c-vqa → 4d-vqa		
****4d-vqa****	S5.4a	Visual QA (Iteration 4)	P5 Implementation	`visual\_qa`	Verify rendered UI against the design spec	`artifacts/4d-vqa/visual\_qa-output.md`	4d → 4e		
****4e****	S5.5	Implementation Iteration 5	P5 Implementation	`implement`, `devops`, `code-review`, `validate`	Builds the code from the approved design, architecture, and requirements / Devops / Code Review agent (Stage 5) / Senior QA Engineer	`artifacts/4e/implement-output.md` `artifacts/4e/devops-output.md` `artifacts/4e/code-review-output.md` `artifacts/4e/validate-output.md`	4d-vqa → 4e-vqa		

| ****4e-vqa**** | S5.5a | Visual QA (Iteration 5) | P5 Implementation | `visual\_qa` | Verify rendered UI against the design spec | `artifacts/4e-vqa/visual\_qa-output.md` | 4e → 4f | | |

| ****4f**** | S5.6 | Implementation Iteration 6 | P5 Implementation | `implement`, `devops`, `code-review`, `validate` | Builds the code from the approved design, architecture, and requirements / Devops / Code Review agent (Stage 5) / Senior QA Engineer | `artifacts/4f/implement-output.md`
`artifacts/4f/devops-output.md` `artifacts/4f/code-review-output.md` `artifacts/4f/validate-output.md` | 4e-vqa → 4f-vqa | | |

| ****4f-vqa**** | S5.6a | Visual QA (Iteration 6) | P5 Implementation | `visual\_qa` | Verify rendered UI against the design spec | `artifacts/4f-vqa/visual\_qa-output.md` | 4f → 5 | | |

| ****5**** | S6.1 | Security Scan | P6 Verification | `security` | "Security analysis with threat modeling, OWASP checks, and vulnerability assessment" | `artifacts/5/security-output.md` | 4f-vqa → 6 | | AG-security-critical |

| ****6**** | S6.2 | NFR Tests | P6 Verification | `validate` | Senior QA Engineer | `artifacts/6/validate-output.md` | 5 → 7 | | |

| ****7**** | S6.3 | Full Test Suite | P6 Verification | `validate` | Senior QA Engineer | `artifacts/7/validate-output.md` | 6 → 8 | | |

| ****8**** | S7.1 | Document | P7 Delivery | `document` | Generates comprehensive documentation for the completed software product including README, user guides, API documentation, architecture diagrams, and | `artifacts/8/document-output.md` `docs/\* (README / guides / API)` | 7 → 9 | | |

| ****9**** | S7.2 | Package | P7 Delivery | `package` | Builds platform-specific packages, installers, and artifacts for the completed software product including BOM, security audits, licenses, and cross-pl | `artifacts/9/package-output.md` `dist/\* (installers / packages)` | 8 → 10 | | |

| ****10**** | S7.3 | Pre-Production | P7 Delivery | `devops`, `orchestrator` | Devops / Project Coordinator (judgment plane) | `artifacts/10/devops-output.md` `artifacts/10/orchestrator-output.md` | 9 → 10a | | AG-deploy |

| ****10a**** | S7.4 | QA Go/No-Go | P7 Delivery | `validate` | Senior QA Engineer | `artifacts/10a/validate-output.md` | 10 → 11 | | |

| ****11**** | S7.5 | Deploy | P7 Delivery | `devops`, `production-deploy` | Devops / Deploys to production with canary/blue-green strategy, feature flags, monitoring, rollback capability. | `artifacts/11/devops-output.md` `artifacts/11/production-deploy-output.md` | 10a → 12 | | AG-deploy |

| ****12**** | S7.6 | Exit | P7 Delivery | `orchestrator` | Project Coordinator (judgment plane) | `artifacts/12/orchestrator-output.md` | 11 → 13 | | |

| ****13**** | S8.1 | Operate & Observability | P8 Operate, Grow & Engage | `observer` | "Future-looking insights, serendipity analysis, and alternative approach suggestions" | `artifacts/13/observer-output.md` | 12 → 13a | yes | AG-operate |

| ****13a**** | S8.2 | Customer Success & Support | P8 Operate, Grow & Engage | `customer-success` | Customer success lead | `artifacts/13a/customer-success-output.md` | 13 → 13b | yes | |

| ****13b**** | S8.3 | Growth, Engagement & Feedback | P8 Operate, Grow & Engage | `growth` | Growth lead | `artifacts/13b/growth-output.md` | 13a → — | yes | |

Agent reference

| Agent | Role | Stages |

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| `architect` | Chooses the tech stack and produces the architecture document (ADRs) from the requirements and design. | 2 |

| `code-review` | Code Review agent (Stage 5) | 4a, 4b, 4c, 4d, 4e, 4f |

| `customer-success` | Customer success lead | 13a |

| `design` | Extracts formal requirements and produces a design doc from the product plan | 1 |

| `design\_critic` | Review the design for quality and completeness. | 1b |

| `devops` | Devops | 4-0, 4a, 4b, 4c, 4d, 4e, 4f, 10, 11 |

| `discovery` | Run structured discovery to shape the product. | 0a |

| `document` | Generates comprehensive documentation for the completed software product including README, user guides, API documentation, architecture diagrams, and | 8 |

| `growth` | Growth lead | 13b |

| `ideation` | Discovery process — explores, challenges, discovers what user actually needs | 0 |

| `implement` | Builds the code from the approved design, architecture, and requirements | 4-0, 4a, 4b, 4c, 4d, 4e, 4f |

| `marketing` | Marketing | 0e |

| `observer` | "Future-looking insights, serendipity analysis, and alternative approach suggestions" | 13 |

| `orchestrator` | Project Coordinator (judgment plane) | 3, 10, 12 |

| `package` | Builds platform-specific packages, installers, and artifacts for the completed software product including BOM, security audits, licenses, and cross-pl | 9 |

| `pricing-strategist` | Pricing & monetization lead | 0d |

| `product-design-spec` | Produce a structured product design specification. | 1a |

| `product-owner` | Works WITH the ideation and discovery agents to distill the idea into product goals, vision, scope, personas and success metrics | 0b |

| `production-deploy` | Deploys to production with canary/blue-green strategy, feature flags, monitoring, rollback capability. | 11 |

| `researcher` | Conducts thorough research on topics using web search, academic sources, and industry knowledge. | 0c, 1d |

| `security` | "Security analysis with threat modeling, OWASP checks, and vulnerability assessment" | 5 |

| `ux-ia` | Define UX and information architecture + design tokens. | 1c |

| `validate` | Senior QA Engineer | 3a, 4a, 4b, 4c, 4d, 4e, 4f, 6, 7, 10a |

| `visual\_qa` | Verify rendered UI against the design spec | 4a-vqa, 4b-vqa, 4c-vqa, 4d-vqa, 4e-vqa, 4f-vqa |

Notes

- **Optional stages** (`0b`, `0c`, `0d`, `0e`, `13`, `13a`, `13b`) are enabled or disabled per project by `core/pipeline\_tailoring.py` (idea-signal based) and recorded in `pipeline-plan.json` (`enabled\_optional` / `disabled\_optional`).
- **Display sequence vs canonical id:** `pipeline-plan.json.display` maps a runtime `display\_seq` (e.g. `S2.1`) to a `canonical\_id` (e.g. `S3.1`) because the visible numbering shifts when optional stages are toggled. The canonical id is the stable reference.
- **Capability generators vs stages:** some outputs (`marketing-strategy/\*`, `onboarding/\*`, `presentations/\*`, `videos/\*`, `product-plan.json`) are produced by capability modules (`core/marketing.py`, `core/customer\_onboarding.py`, `core/presentation\_generator.py`, `core/product\_plan.py`) run as stage hooks — independent of stages `0b–0e`.
- **Spec ids** (FR/NFR/US + local families) are parsed by the single canonical query `core/id\_index.py`; id families live in `config/spec-id-families.json`. Feature ids are a running number across features (no hard cap): F-1 `FR-1..n`, F-2 `n+1..`, via `core/id\_index.renumber\_global`.
- **Single-run guard:** `core/run\_guard.py` stops any live instance before a start/restart/resume/continue and takes the project lock (`products/.locks/`).
- **run_id linkage:** approvals carry `run\_id`; `core/run\_state.py` records run attempts in `pipeline-runs.json` and reconciles already-approved stages on resume (preserve-first — user data is never cleared).